

SEC. 2 PROBLEM FORMULATION

1. PROBLEM FORMULATION

Groups and objectives. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a feature space and $\mathcal{Y}$ a label space, and let the population be partitioned into N groups indexed by $i \in [N]$ (demographic categories, devices, environments) with *prevalences* $\pi \in \Delta^{N-1}$, $\pi_i = \mathbb{P}[\text{group } i]$. Group i carries a distribution $\mathcal{D}_i$ on $\mathcal{X} \times \mathcal{Y}$ and the risk

$$f_i(\theta) \triangleq \mathbb{E}_{(X,Y) \sim \mathcal{D}_i} [\ell(m(X; \theta), Y)], \quad \theta \in \mathcal{C} \subseteq \mathbb{R}^{d'},$$

where $m(\cdot; \theta)$ is the predictor and ℓ the loss. Empirical risk minimization weights the groups by prevalence, $F_\pi = \sum_i \pi_i f_i$, and a majority group therefore dominates the trained model. The remedy is to fix an *importance distribution* $p \in \Delta^{N-1}$ and minimize

$$F_p(\theta) \triangleq \sum_{i=1}^N p_i f_i(\theta), \quad \theta^* \in \arg \min_{\theta \in \mathcal{C}} F_p(\theta). \quad (1)$$

Uniform p is the category-uniform objective, $p_i \propto 1/\pi_i$ the inverse-prevalence one, and p may equally encode a fairness constraint or be imposed externally [4, 5]. Throughout,

Assumption 1. $\mathcal{C}$ is convex and compact with diameter D ; each f_i is convex on $\mathcal{C}$ and $0 \leq f_i \leq M$; and every stochastic subgradient $g_i(\theta)$ of f_i returned by the sampler satisfies $\mathbb{E} \|g_i(\theta)\|^2 \leq G^2$ on $\mathcal{C}$.

These are the assumptions of [7, 6] restated for groups; they place no restriction on p or on π .

Prevalence versus importance. Minibatch SGD does not see p . At step t the sampler draws a batch B^t of b examples; write

$$A^t \triangleq \{i \in [N] : B^t \text{ contains an example from group } i\}$$

for the *active set* of groups, let $\{A_j\}_{j \in [M]}$ enumerate its support, and let $q_j \triangleq \mathbb{P}[A^t = A_j]$ be the *composition law*. The observation rate of group i is $r_i \triangleq \mathbb{P}[i \in A^t] = \sum_{j: i \in A_j} q_j$; under i.i.d. sampling $r_i = 1 - (1 - \pi_i)^b$, so a group whose prevalence is below the resolution of the batch is absent from most batches whatever p says. This is federated learning under partial participation: A^t is the set of participating clients, q the availability distribution, p the importance distribution [6, 7]. We treat q as known; otherwise it is estimated offline by simulating the sampler, as in our experiments.

What group-blind SGD minimizes. The group-blind update averages the per-example subgradients of B^t , which assigns group $i \in A^t$ the weight $\hat{\pi}_i^t$, its share of the batch, and assigns absent groups zero. Averaging first over the batch given A^t and then over $A^t \sim q$ is the hierarchical experiment of [7, Eq. 4], and it names the function SGD is in fact minimizing.

Proposition 2 (Surrogate objective). *Under Assumption 1, group-blind minibatch SGD is a stochastic subgradient method on $F_{\tilde{p}}$ with*

$$\tilde{p}_i = \sum_{j: i \in A_j} q_j \mathbb{E} [\hat{\pi}_i^t | A^t = A_j], \quad (2)$$

a probability vector. For i.i.d. sampling $\tilde{p} = \pi$; for uniform weighting within the batch $\tilde{p}_i = \sum_{j: i \in A_j} q_j / |A_j|$, the surrogate of agnostic *FedAvg* [6, Eq. 2]. Consequently, for every $\theta \in \mathcal{C}$,

$$|F_p(\theta) - F_{\tilde{p}}(\theta)| \leq M \|p - \tilde{p}\|_1, \quad (3)$$

and any method that minimizes $F_{\tilde{p}}$ exactly incurs the excess $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq 2M \|p - \tilde{p}\|_1$ on (1), which does not vanish with T .

Proof. Conditionally on $A^t = A_j$ the within-group averages are unbiased for ∂f_i , so $\mathbb{E}[\text{batch subgradient} | A^t = A_j] = \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t | A_j] \partial f_i(\theta)$; averaging over q gives $\sum_i \tilde{p}_i \partial f_i = \partial F_{\tilde{p}}$, and $\sum_i \tilde{p}_i = \sum_j q_j \sum_{i \in A_j} \mathbb{E}[\hat{\pi}_i^t | A_j] = 1$. Both bounds follow from $0 \leq f_i \leq M$ and the two-point inequality $F_p(\theta_{\tilde{p}}^*) - F_p(\theta^*) \leq [F_p - F_{\tilde{p}}](\theta_{\tilde{p}}^*) + [F_{\tilde{p}} - F_p](\theta^*)$. $\square$

Proposition 2 says that the choice of p in (1) is decorative unless the aggregation step is changed: the iterates converge, at the usual rate, to the minimizer of a function whose weights are induced by the sampler rather than chosen, and the bias $p - \tilde{p}$ survives every step-size schedule. Rebalancing the sampler is only a partial fix: $\tilde{p}_i \leq r_i$ always, so a group with $p_i > r_i$ cannot be sampled into enough batches however the sampler is tilted, while upscaling the examples that do appear reintroduces an amplitude distortion that destabilizes the iterates [6]. The question is the one posed for federated aggregation in [6], read on batches:

Can the per-batch group weights be chosen so that SGD driven by the composition law q converges for the objective defined by p ?

The next section answers it, and characterizes exactly when the answer is yes.